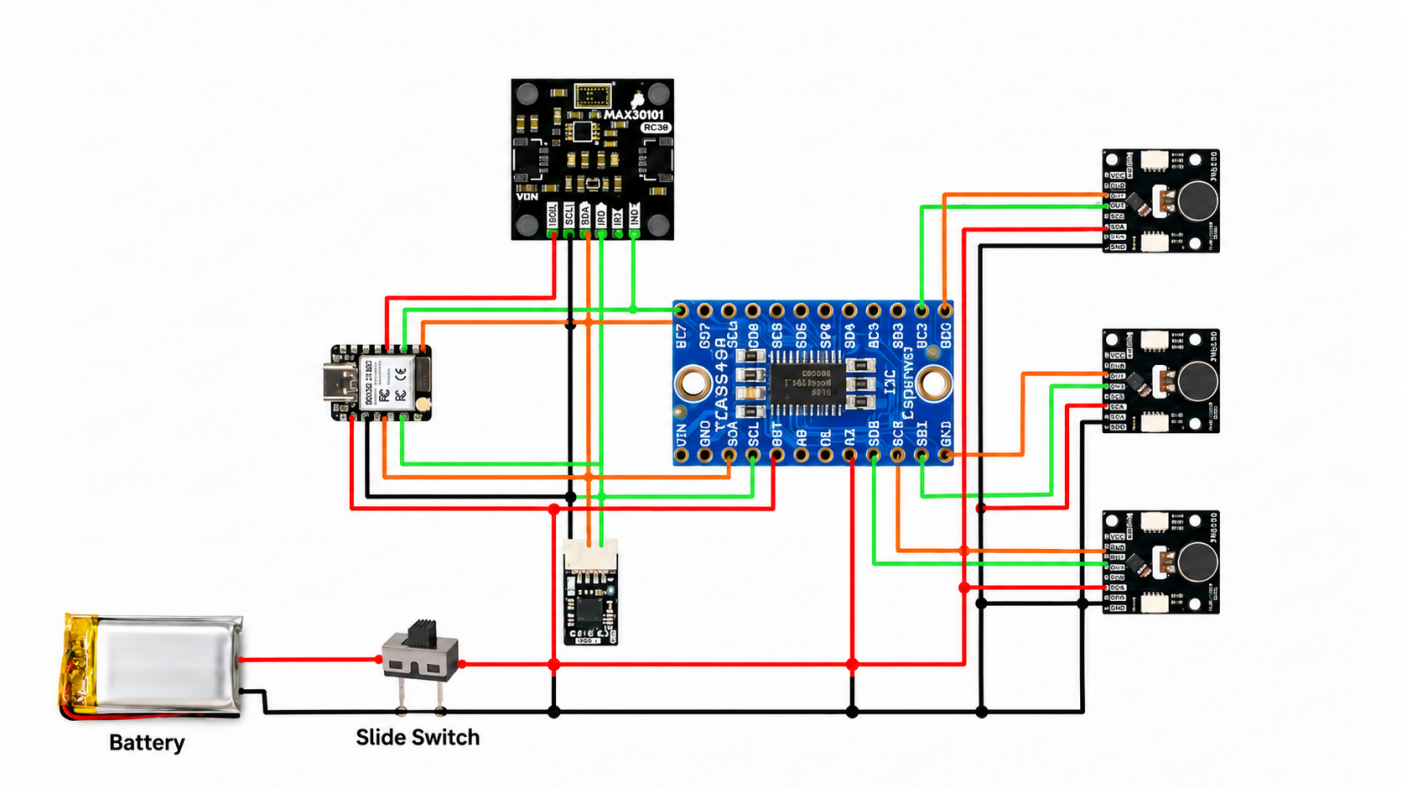
Sutra Hardware Schematic

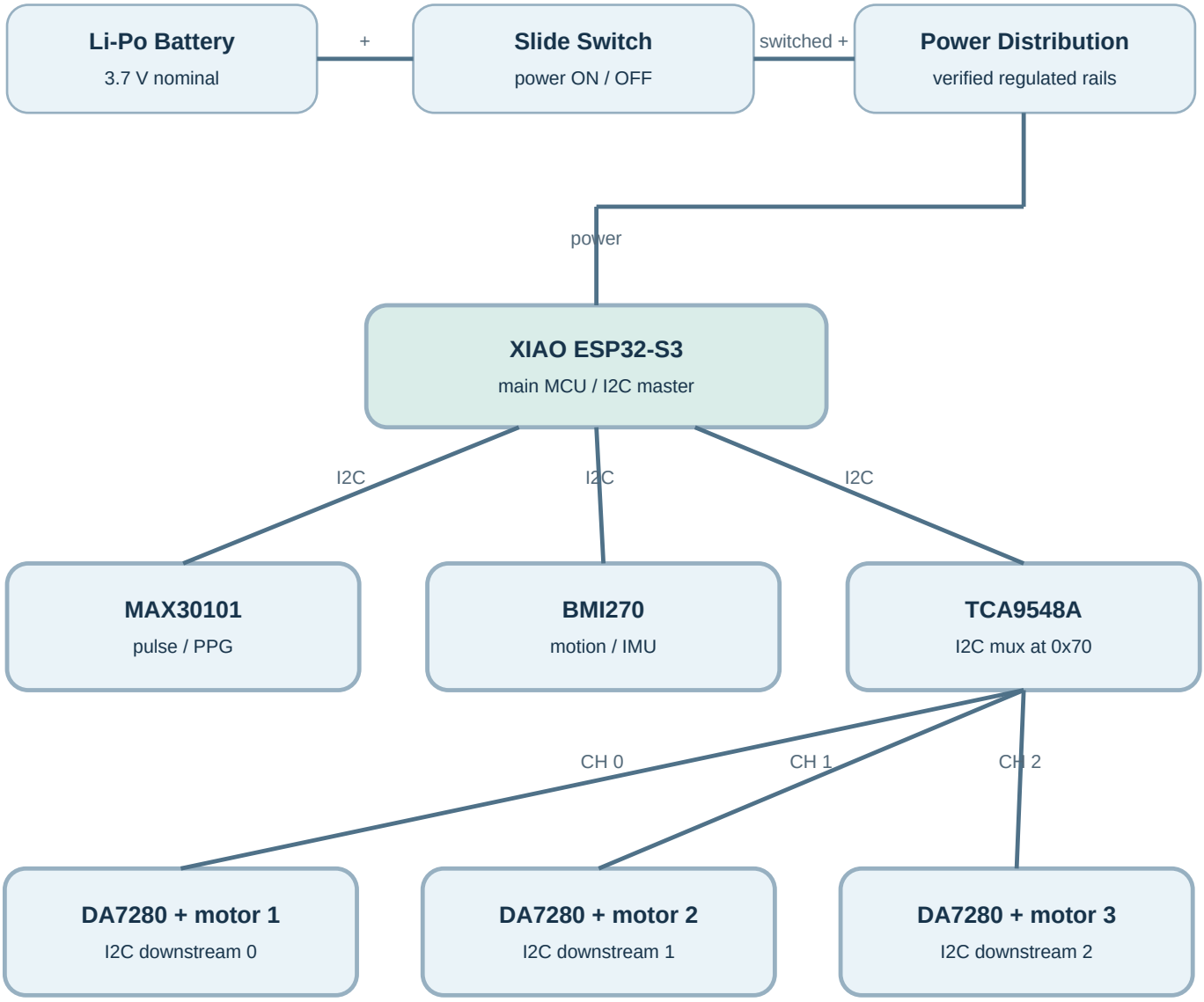


Components shown

1. Seeed XIAO ESP32-S3 microcontroller
2. MAX30101 optical pulse sensor
3. BMI270 IMU module (small lower board, assumed)
4. TCA9548A I2C multiplexer
5. Three DA7280 haptic driver / motor modules
6. Lithium-polymer battery and slide ON/OFF switch

Illustration is conceptual: some printed pin labels and power connections may not be accurate.

Simplified Electrical Connections



Basic wiring summary

Power: Battery (+) -> slide switch -> suitable regulated power rails; battery (-) -> common GND.

Communication: MCU SDA / SCL connect to MAX30101, BMI270 and TCA9548A input bus.

Haptics: TCA9548A channels 0, 1 and 2 connect to the three DA7280 modules.

Motor outputs are driven by the respective DA7280 boards, not MCU GPIO pins.

Before wiring hardware

Confirm each board voltage range and I2C pull-ups. A raw Li-Po cell is 4.2 V when fully charged.

Use appropriate battery protection/charging and regulation; verify switch current rating and pinout.